

2025 Linear Algebra and Calculus Final Exam

- ① A particle moves through $s(t) = t^2 e^{-t}$, $t \geq 0$.

The particle is at rest when $s'(t) = 0$. So, $s'(t) = 2te^{-t} - t^2 e^{-t} = e^{-t}(2t - t^2) = 0$, which only happens if $-t^2 + 2t = 0 \rightarrow t(-t + 2) = 0 \rightarrow t = 0$ and $t = 2$. If the acceleration is given by $s''(t)$, if $s''(t) = 0$, there is no acceleration, so if $s''(t) = -e^{-t}(2t - t^2) + e^{-t}(2 - 2t) = e^{-t}(-2t + t^2 + 2 - 2t) = e^{-t}(t^2 - 4t + 2)$, so $s''(1) = e^{-1}(1 - 4 + 2) = -\frac{1}{e} < 0$, so the particle speeds down (or brakes).

- ② $f(x, y, z) = x^2 y + y z^3 - 4xz$.

$$a) \nabla f(x, y, z) = \begin{bmatrix} \frac{\partial f}{\partial x}(x, y, z) \\ \frac{\partial f}{\partial y}(x, y, z) \\ \frac{\partial f}{\partial z}(x, y, z) \end{bmatrix} = \begin{bmatrix} 2xy - 4z \\ x^2 + z^3 \\ 3zy - 4x \end{bmatrix}$$

$$b) D_v f(P) = v \cdot \nabla f(P) = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 4+4 \\ 1+(-1) \\ 6-4 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 8 \\ 0 \\ 2 \end{bmatrix} = 18.$$

- ③ $f(x) = \begin{cases} \frac{Kx^2 - 9K}{x-3} & , x < 3 \\ 2x^2 - 16 & , x \geq 3 \end{cases}$

$$\lim_{x \rightarrow 3} \frac{Kx^2 - 9K}{x-3} = \lim_{x \rightarrow 3} \frac{K(x^2 - 9)}{x-3} = \lim_{x \rightarrow 3} \frac{K(x-3)(x+3)}{x-3} = \lim_{x \rightarrow 3} K(x+3) = 6K$$

Since $f(3) = 2$, f is continuous if $\lim_{x \rightarrow 3} f(x) = f(3)$, so $6K = 2 \rightarrow K = \frac{1}{3}$

- ④ Prove existence of solutions for $x^3 + \sin x = 3x$ in $[-2, -1]$.

It's equivalent to $x^3 - 3x + \sin x = 0$

Let's call $f(x) = x^3 - 3x + \sin x$. $f(-2) = -8 + 6 + \sin(-2) = -2 + \sin(-2) < 0$, since $\sin(-2)$ is at most 1. $f(-1) = -1 + 3 + \sin(-1) = 2 + \sin(-1) > 0$, since $\sin(-1)$ is at least -1. Since f is a sum of continuous functions, it is continuous. By Bolzano's Theorem, since $f(-2) \cdot f(-1) < 0$, $\exists c \in (-2, -1)$ such that $f(c) = 0$, which is a solution to the original equation.

- ⑤ $T(x, y, z) = (x+y, 2x-z, y+z)$

a) Standard matrix of T .

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

b) Basis for $\text{Ker } T$.

Since T is a matrix map, $\text{Ker } T = \text{Nul } A$, so let's solve $Ax = 0$.

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The system has unique solution, so

$$\text{Ker } T = \text{Nul } A = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

c) Basis for $\text{Im } T$.

Since T is a matrix map, $\text{Im } T = \text{Col } A$. Since $\dim \text{Ker } T = 0$, $\dim \text{Im } T = 3 - \dim \text{Ker } T = 3$, so a basis for $\text{Im } T$ is $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$.

⑥ a) Compute the inverse of $A = \frac{1}{8} \begin{bmatrix} 4 & -2 & -2 \\ -2 & 5 & 1 \\ -2 & 1 & 5 \end{bmatrix}$

To find the inverse of A , we do $[A | I] \sim [I | A^{-1}]$.

$$\begin{bmatrix} 4/8 & -2/8 & -2/8 & 1 & 0 & 0 \\ -2/8 & 5/8 & 1/8 & 0 & 1 & 0 \\ -2/8 & 1/8 & 5/8 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 4/8 & -2/8 & -2/8 & 1 & 0 & 0 \\ 0 & 4/8 & 0 & 1/2 & 1 & 0 \\ 0 & 0 & 4/8 & 1/2 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1/2 & -1/2 & 2 & 0 & 0 \\ 0 & 1 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1/2 & 0 & 5/2 & 0 & 1 \\ 0 & 1 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 3 & 1 & 1 \\ 0 & 1 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 & 2 \end{bmatrix}$$

A^{-1}

b) Assume $B \neq 0$ is a square matrix such that $B^2 = 0$. Show that B is not invertible.

Let's assume it is, so there exists a matrix B^{-1} such that $B^{-1}B = I$. Multiplying on the left (or right) of $B^2 = 0$, we see:

$$\begin{aligned} B^2 = 0 &\iff B^{-1}B^2 = B^{-1}0 \iff B^{-1}(BB) = 0 \iff (B^{-1}B) \cdot B = 0 \iff I \cdot B = 0 \\ &\implies B = 0, \text{ which contradicts the assumption that } B \neq 0. \text{ So } B \text{ cannot be invertible.} \end{aligned}$$

⑦ $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0, z - y = 0\}$

a) Show that W is a subspace of $\mathbb{R}^3$.

For that, we need to prove three things:

$$\begin{aligned} - & 0 + 0 + 0 = 0, z - y = 0 \implies 0 \in W \\ - & u, v \in W, (u_1 + v_1) + (u_2 + v_2) + (u_3 + v_3) = \\ & = (u_1 + u_2 + u_3) + (v_1 + v_2 + v_3) = 0 + 0 = 0 \end{aligned}$$

$$z(u_1 + v_1) - (u_2 + v_2) = zu_1 + zv_1 - u_2 - v_2 = zu_1 - u_2 + zv_1 - v_2 = 0 + 0 = 0$$

$$\implies u + v \in W$$

$$- u \in W, c \in \mathbb{R}, cu_1 + cu_2 + cu_3 = c(u_1 + u_2 + u_3) = c \cdot 0 = 0$$

$$z(cu_1) - cu_2 = c(zu_1 - u_2) = c \cdot 0 = 0$$

$$\implies cu \in W$$

b) Find a basis for W .

$$\begin{cases} x+y+z=0 \\ 2x-y=0 \end{cases} \text{ Augmented matrix } \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & -1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & -3 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 2/3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1/3 & 0 \\ 0 & 1 & 2/3 & 0 \end{bmatrix}$$

$$\Rightarrow \text{Basis: } \left\{ \begin{bmatrix} -1/3 \\ -2/3 \\ 1 \end{bmatrix} \right\}.$$

$$\begin{aligned} x_1 &= -\frac{1}{3}x_3 \\ x_2 &= -\frac{2}{3}x_3 \end{aligned} \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -1/3 \\ -2/3 \\ 1 \end{bmatrix} \right.$$

c) Dimension of W .

Since the basis has one vector, the dimension is 1.

⑧ Let $A = \begin{bmatrix} 4 & 1 & -2 \\ 0 & 3 & 0 \\ 2 & 1 & 0 \end{bmatrix}$

a) Find the eigenvalues of A .

$$\text{For that we solve for } \lambda, \det(A - \lambda I) = 0 \rightarrow (3-\lambda) \begin{vmatrix} 4-\lambda & -2 \\ 2 & -\lambda \end{vmatrix} = (3-\lambda)(-\lambda(4-\lambda)+4) = (3-\lambda)(\lambda^2-4\lambda+4) = 0$$

$$\text{So } \lambda = 3 \text{ (mult. 1), and } \lambda = 2 \text{ (mult. 2).}$$

$$\lambda = 3 \quad \lambda = \frac{4 \pm \sqrt{16-16}}{2} = 2$$

b) Basis for eigenspaces.

We must solve $(A - \lambda I)x = 0$ for each eigenvalue.

For $\lambda = 3$:

$$A - 3I = \begin{bmatrix} 1 & 1 & -2 \\ 0 & 0 & 0 \\ 2 & 1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{aligned} x_1 &= x_3 \\ x_2 &= x_3 \\ x_3 &= x_3 \end{aligned} \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right. \rightarrow \text{Basis: } \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

For $\lambda = 2$

$$A - 2I = \begin{bmatrix} 2 & 1 & -2 \\ 0 & 1 & 0 \\ 2 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/2 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{aligned} x_1 &= x_3 \\ x_2 &= 0 \\ x_3 &= x_3 \end{aligned} \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right. \rightarrow \text{Basis: } \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

c) Is A diagonalizable?

No, since $\lambda = 2$ has multiplicity 2 but the dimension of its eigenspace is 1.

⑨ $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$

a) Orthogonal basis.

Since $\{v_1, v_2, v_3\}$ is linearly independent, we can apply the Gram-Schmidt process.

$$u_1 = v_1$$

$$u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix}$$

$$u_3 = v_3 - \frac{v_3 \cdot u_1}{u_1 \cdot u_1} u_1 - \frac{v_3 \cdot u_2}{u_2 \cdot u_2} u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix} \rightarrow \text{Basis: } \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix}, \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix} \right\}$$

b) Orthogonal projection of $y = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$ onto $\text{Span}\{v_1, v_2, v_3\}$.

Since $y = v_1 + v_2 + v_3$, the projection is y itself.

10) True or false?

a) If A is invertible, $\det A^{-1} = \frac{1}{\det A}$.

True. $A^{-1}A = I \rightarrow \det(A^{-1}A) = \det I \rightarrow \det A^{-1} \det A = 1 \rightarrow \det A^{-1} = \frac{1}{\det A}$.

b) $(A+B)^2 = A^2 + 2AB + B^2$.

False, because matrix product is in general non-commutative, $AB \neq BA$, so $(A+B)^2 = A^2 + AB + BA + B^2$.

c) $\{1, t^2, t^4\}$ is a basis for P_4 .

False. $t^3 \in P_4$ but it can't be combined from $1, t^2$ and t^4 .